

3. Ans: B

$$\text{Total momentum} = 2Mv - Mv = Mv$$

Option A is a correct statement (so don't choose it). The spheres cannot come to rest at the same time as that will give a total momentum of 0 and not Mv .

Option C is a correct statement. For total momentum to be conserved, the change in momentum of A and B must be equal and opposite.

Option D is a correct statement. Newton's 3rd law.

